

NANO-MAMBA U-NET: A SPATIO-TEMPORAL
FRAMEWORK FOR CARDIAC MOTION ANALYSIS

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**NANO-MAMBA U-NET: A SPATIO-TEMPORAL
FRAMEWORK FOR CARDIAC MOTION ANALYSIS**

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NANO-MAMBA U-NET: A SPATIO-TEMPORAL FRAMEWORK FOR CARDIAC MOTION ANALYSIS

ABSTRACT

Cardiac Magnetic Resonance (CMR) imaging is widely used for quantitative assessment of cardiac anatomy and function. In automated cardiac analysis, accurate segmentation of the right ventricle (RV), myocardium (MYO), and left ventricle (LV) is an important prerequisite for deriving functional measurements such as end-diastolic volume, end-systolic volume, myocardial mass, and ejection fraction. Although U-Net-style convolutional neural networks remain strong foundations for medical image segmentation, practical deployment also requires attention to model size, inference cost, and reproducible validation. This thesis investigates Nano-Mamba U-Net, a lightweight Mamba-inspired 3D U-Net architecture for cardiac MRI segmentation. The model keeps the encoder-decoder structure and skip connections of U-Net, while inserting a compact sequence-processing bottleneck at the compressed latent representation. This bottleneck serializes 3D feature maps into a one-dimensional sequence, applies a lightweight gated sequence module inspired by Mamba-style state space modeling, and reshapes the output back into volumetric form. The implementation is therefore described as Mamba-inspired rather than as a full reproduction of the original hardware-aware selective scan algorithm.

The proposed model was evaluated on the Automated Cardiac Diagnosis Challenge (ACDC) dataset using a deterministic patient-level validation protocol. The final experiment used 80 training patients and 20 held-out validation patients, corresponding to 160 training cases and 40 validation cases. All compared models were trained under the same preprocessing and validation procedure, and the best checkpoint for each model was selected by validation mean Dice. Nano-Mamba U-Net achieved a validation mean Dice Similarity

Coefficient (DSC) of 84.78% with 1.46 million trainable parameters. It improved over the 3D U-Net baseline, which achieved 80.83% mean DSC with 4.81 million parameters. However, SegResNet16 achieved the highest mean DSC of 86.70%, and the No-Mamba ablation achieved 85.64%. These results show that Nano-Mamba U-Net should not be claimed as the most accurate model by Dice. Instead, its main contribution is an accuracy-efficiency trade-off: it provides competitive held-out validation performance while using the smallest parameter count among the evaluated models. The thesis concludes that lightweight sequence-inspired bottlenecks are promising for cardiac MRI segmentation, but further work is required for official test-set evaluation, pathology-specific analysis, external validation, and full 4D cardiac motion modeling.

Keywords: Cardiac MRI Segmentation, Deep Learning, Mamba-Inspired Architecture, State Space Models, Accuracy-Efficiency Trade-off.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

1.1.1 Clinical Significance of Cardiovascular Diseases

Cardiovascular diseases (CVDs) remain one of the most important global causes of mortality. According to the World Health Organization (WHO), CVDs caused an estimated 19.8 million deaths in 2022, corresponding to approximately 32% of all global deaths (World Health Organization, 2025). This figure is not used in this thesis as a direct claim about the performance of any computational model. Rather, it establishes the clinical context in which efficient and reproducible cardiac image analysis tools are valuable. When the global disease burden is high, even small improvements in the reliability or scalability of cardiac assessment workflows can have practical importance.

Cardiac Magnetic Resonance (CMR) imaging is widely used for non-invasive assessment of cardiac anatomy and function (Bernard et al., 2018). From CMR images, clinicians can derive functional indices such as Ejection Fraction (EF), End-Diastolic Volume (EDV), End-Systolic Volume (ESV), and Myocardial Mass. These measurements depend on accurate delineation of the Right Ventricle (RV), Myocardium (MYO), and Left Ventricle (LV). Therefore, automated segmentation of these structures is a central step in computer-assisted cardiac analysis. In this project, segmentation is treated as the computational foundation for cardiac motion-related analysis, because ED and ES phase segmentation enables estimation of cardiac volume changes across the cardiac cycle.

The clinical relevance of this task comes from the relationship between segmentation masks and quantitative cardiac indices. If V_{ED} denotes the ventricular volume at end-diastole and V_{ES} denotes the volume at end-systole, ejection fraction can be expressed

as:

$$EF = \frac{V_{ED} - V_{ES}}{V_{ED}} \times 100\%. \quad (1.1)$$

This equation shows why segmentation errors matter. A small boundary error can change the estimated cavity volume, and the resulting volume error can propagate into EF. Thus, the segmentation model is not merely producing visually plausible masks; it supports quantitative measurements used in cardiac functional assessment.

1.1.2 The Bottleneck of Manual Delineation and Traditional Methods

Manual delineation of cardiac structures requires expert knowledge and is time-consuming, particularly when a cine CMR study contains multiple slices and cardiac phases. The ACDC challenge paper emphasizes that accurate segmentation of RV, MYO, and LV is essential for cardiac diagnosis and that manual annotation can be affected by inter-observer variability (Bernard et al., 2018). Instead of relying on an unsupported fixed time estimate, this thesis treats manual annotation as a practical bottleneck because it is labor-intensive, repetitive, and difficult to scale across large datasets.

Manual segmentation is also not purely mechanical. The myocardium has an annular shape and may be difficult to separate from papillary muscles. The RV has a thin wall and variable geometry. The basal and apical slices often contain ambiguous anatomy because the visible structures gradually appear or disappear across the slice direction. These difficulties mean that even expert annotations may vary. Automated segmentation is therefore attractive not because it removes the need for clinical expertise, but because it can provide a consistent computational starting point for measurement and review.

Traditional image processing methods, including thresholding, active contours, level sets, and graph-cut algorithms, were historically applied to medical image segmentation. However, these methods rely heavily on hand-crafted features, manual initialization, or

assumptions about image intensity distributions. In CMR images, boundary ambiguity, partial volume effects, papillary muscles, and scanner-dependent intensity variations make fully automated segmentation challenging (Peng et al., 2020). These limitations motivated the transition toward deep learning methods that learn task-specific representations from annotated data.

1.1.3 Deep Learning for Cardiac Segmentation

Deep learning shifted medical image segmentation from hand-crafted feature engineering toward data-driven representation learning. Fully Convolutional Networks (FCNs) introduced dense pixel-wise prediction for semantic segmentation (Long et al., 2015), while U-Net became a standard architecture for biomedical image segmentation due to its encoder-decoder design and skip connections (Ronneberger et al., 2015). For volumetric medical images, 3D U-Net extended this idea to three-dimensional convolutional processing (Çiçek et al., 2016). These architectures are important foundations for this thesis, and this project does not claim them as original contributions.

Despite their success, convolutional networks still rely on local kernels. A $3 \times 3 \times 3$ convolution extracts information from a local voxel neighbourhood. Stacking several convolutional layers increases the effective receptive field, but the operation remains locally constructed. In cardiac MRI, local detail and wider anatomical context are both important. The model must distinguish the myocardium from the blood pool at local boundaries while also maintaining enough contextual awareness to understand the relative positions of RV, MYO, and LV.

Transformer-based architectures address long-range interactions through self-attention (Dosovitskiy et al., 2020; Vaswani et al., 2017). Medical segmentation models such as UNETR and Swin UNETR demonstrate the usefulness of global or window-based token interactions in volumetric segmentation tasks (Hatamizadeh, Nath, et al., 2022; Hatamizadeh,

Tang, et al., 2022). However, self-attention can increase memory and computational requirements, which motivates research into more efficient sequence modeling alternatives. This thesis does not argue that Transformers are unsuitable for medical imaging. Instead, it investigates whether a lighter sequence-inspired bottleneck can be useful when the hardware environment is constrained.

1.1.4 Motivation for a Mamba-Inspired Bottleneck

Selective State Space Models (SSMs), including Mamba, have recently attracted attention because they can model long sequences with linear-time complexity (Gu & Dao, 2023). This thesis investigates whether a lightweight Mamba-inspired sequence bottleneck can be integrated into a 3D U-Net-style architecture for cardiac MRI segmentation. The implemented model serializes compressed 3D volumetric features into a one-dimensional token sequence, applies a lightweight gated sequence module inspired by Mamba, and reshapes the result back into volumetric form.

The motivation is based on the accuracy-efficiency trade-off. A model with the highest Dice score is not necessarily the most suitable model for all environments if it requires more memory, more parameters, or longer inference time. Conversely, a very small model is not useful if it fails to segment clinically relevant structures. This thesis therefore evaluates Nano-Mamba U-Net not only by segmentation accuracy but also by parameter count and inference speed. The research question is whether a compact sequence-inspired bottleneck can produce competitive validation performance while maintaining a small model size.

The thesis title refers to cardiac motion analysis because segmentation of end-diastolic and end-systolic cardiac frames is a prerequisite for computing motion-related cardiac function indicators such as EDV, ESV, and EF. The implemented scope of this project is not dense motion-field estimation or optical-flow tracking. Instead, it focuses on segmen-

tation across cardiac phases as a foundation for functional cardiac assessment. This distinction is important for scientific accuracy: the project supports cardiac motion analysis through phase-based anatomical segmentation, but it does not directly estimate pixel-wise displacement fields.

1.2 Scope of the Study

The scope of this research is defined as follows:

- **Dataset Boundary:** The empirical evaluation uses the Automated Cardiac Diagnosis Challenge (ACDC) dataset (Bernard et al., 2018). The segmentation targets are RV, MYO, and LV.
- **Task Boundary:** The project focuses on 3D cardiac MRI segmentation of labelled cardiac frames. Motion analysis is considered indirectly through segmentation of clinically meaningful cardiac phases rather than through explicit motion-vector estimation.
- **Implementation Boundary:** The proposed Nano-Mamba U-Net uses a Mamba-inspired bottleneck implemented in PyTorch. It should not be interpreted as a full reproduction of the original hardware-aware selective scan implementation.
- **Evaluation Boundary:** The main reported results are based on a deterministic patient-level 80/20 validation split of the ACDC training cohort. The study does not claim performance on the hidden official ACDC test set.
- **Hardware Boundary:** Training, validation, and computational profiling are conducted on a single consumer-grade NVIDIA RTX 4060 GPU with 8GB VRAM.

These boundaries are deliberately stated to avoid over-claiming. The goal of the thesis is to produce a rigorous and reproducible project-level investigation, not to claim clini-

cal deployment readiness or universal superiority over all existing cardiac segmentation methods.

1.3 Organization of the Thesis

This thesis is organized into eight chapters. Chapter 1 introduces the clinical and computational motivation. Chapter 2 defines the research problem. Chapter 3 states the objectives and research questions. Chapter 4 reviews relevant work in cardiac image segmentation, CNNs, attention models, residual networks, and state space models. Chapter 5 describes the dataset, preprocessing pipeline, architecture, training protocol, and evaluation method. Chapter 6 reports the validation results. Chapter 7 discusses the findings and limitations. Chapter 8 concludes the thesis and outlines future work.

CHAPTER 2: RESEARCH PROBLEM

Automated cardiac MRI segmentation is not only an accuracy problem. A clinically useful model must also be computationally practical and robust across different cardiac anatomies. This thesis focuses on the gap between accurate volumetric segmentation and lightweight implementation. The problem is framed around three connected issues: local context modeling, computational cost, and anatomical variability.

2.1 Local Context Modeling in Lightweight 3D CNNs

3D CNNs are effective for volumetric segmentation because they process spatial neighborhoods directly. However, a standard $3 \times 3 \times 3$ convolution only aggregates local voxel information at each layer. Although deeper networks can enlarge the effective receptive field, lightweight models with limited depth and channel capacity may still struggle to capture broader contextual relationships across the heart. This limitation is important in CMR segmentation because RV, MYO, and LV boundaries can be ambiguous at basal and apical slices.

The issue is not that convolution is ineffective. Convolution is one of the reasons U-Net-style models are successful in medical segmentation. The problem is that a lightweight model must compress information aggressively to remain small. When the deepest representation is too narrow or too local, the model may lose broader anatomical relationships. A useful bottleneck should therefore preserve compactness while allowing information exchange across the compressed spatial representation. This is the specific architectural motivation for inserting a sequence-processing block at the bottleneck rather than replacing the entire U-Net with a large attention model.

2.2 Computational Cost of Larger Segmentation Models

Stronger segmentation architectures often increase depth, width, attention operations, or residual blocks to improve representation capacity. This can improve accuracy but also increases parameters, memory usage, and inference cost. For resource-constrained clinical, educational, or prototype environments, a model with lower parameter count and acceptable accuracy may be preferable to a heavier model with marginally higher Dice scores. Therefore, the central problem is not simply to maximize Dice, but to study the trade-off between segmentation accuracy and computational efficiency.

This thesis uses parameter count and inference speed as supporting metrics because Dice alone does not describe the full engineering value of a model. A model with 1.46M parameters and competitive Dice may be useful even if it is not the highest-accuracy model. Conversely, if a lightweight model achieves lower Dice without meaningful savings, it would not be justified. The final evaluation therefore compares Nano-Mamba U-Net against baselines and ablation models using the same split and training protocol.

2.3 Pathological and Anatomical Variability

The ACDC dataset contains normal subjects and multiple cardiac pathology groups, including dilated cardiomyopathy, hypertrophic cardiomyopathy, myocardial infarction, and abnormal right ventricle cases (Bernard et al., 2018). These cases introduce variation in ventricular size, myocardial thickness, and boundary appearance. A segmentation model must therefore handle both normal cardiac geometry and pathological deformation.

Anatomical variability is particularly important for the RV and MYO. The RV has a complex shape and thinner wall, which makes its segmentation more sensitive to boundary ambiguity. The MYO is a narrow structure, so small absolute boundary errors can cause a noticeable Dice reduction. This explains why class-specific Dice is reported in the results chapter instead of reporting only one aggregate number. The evaluation must

show whether the model performs consistently across RV, MYO, and LV.

2.4 Research Gap

Existing segmentation backbones such as 3D U-Net, Attention U-Net, and SegResNet provide strong baselines, but they represent different points on the accuracy-efficiency spectrum. The research gap addressed in this thesis is whether a compact Mamba-inspired bottleneck can provide competitive validation performance with fewer parameters than conventional or residual alternatives, while remaining feasible on a consumer GPU.

The gap is not a claim that previous methods are inadequate in all settings. SegResNet, for example, remains a strong residual convolutional baseline. The gap is narrower and more appropriate for this project: there is value in testing a compact sequence-inspired bottleneck under a rigorous patient-level validation protocol, especially when earlier exploratory scripts may overestimate performance by evaluating on non-held-out data. This thesis therefore emphasizes reproducibility and honest comparison as much as architectural novelty.

CHAPTER 3: RESEARCH OBJECTIVES

The aim of this research is to design, implement, and evaluate a lightweight Mamba-inspired U-Net architecture for cardiac MRI segmentation. The study emphasizes empirical validation, parameter efficiency, and transparent comparison with established segmentation baselines. The objectives are designed to connect directly to the experiments reported in Chapter 6.

3.1 Specific Objectives

1. To implement a Nano-Mamba U-Net architecture that inserts a lightweight Mamba-inspired sequence bottleneck into a 3D U-Net-style encoder-decoder network.
2. To evaluate the segmentation performance of the proposed model on RV, MYO, and LV using Dice Similarity Coefficient on a deterministic patient-level validation split of the ACDC training cohort.
3. To compare the proposed model with 3D U-Net, Attention U-Net, SegResNet, and ablation variants using the same split, training settings, and validation protocol.
4. To quantify computational efficiency using parameter count, inference speed, and GPU memory profiling where applicable.

The first objective is implemented through the model architecture described in Chapter 5. The second and third objectives are addressed through the quantitative validation table in Chapter 6. The fourth objective is addressed by reporting trainable parameters and inference speed together with Dice. This mapping is included to make the thesis structure explicit and to ensure that the stated objectives are actually answered by the experimental design.

3.2 Research Questions

1. How can compressed 3D cardiac MRI feature maps be serialized and processed by a Mamba-inspired bottleneck within a U-Net-style segmentation architecture?
2. Does the proposed Nano-Mamba U-Net achieve competitive held-out validation Dice compared with 3D U-Net, Attention U-Net, SegResNet, and ablation variants?
3. What accuracy-efficiency trade-off is observed when comparing Nano-Mamba U-Net with heavier or non-Mamba alternatives in terms of Dice score, parameter count, and inference speed?

These research questions are intentionally phrased in a measurable way. The first question is answered by architecture implementation rather than by a numerical score alone. The second question is answered by held-out validation Dice. The third question is answered by jointly interpreting Dice, parameter count, and speed. This structure avoids unsupported claims such as absolute superiority and keeps the thesis aligned with the actual experimental evidence.

CHAPTER 4: LITERATURE REVIEW

This chapter reviews the scientific and technical literature that supports the design choices of this thesis. The purpose is not to claim that the proposed model replaces all existing cardiac segmentation methods. Instead, the review identifies the specific research context in which a lightweight Mamba-inspired U-Net is meaningful: automated cardiac MRI segmentation requires accurate anatomical delineation, but practical models must also consider parameter count, memory usage, and reproducible validation.

4.1 Cardiac MRI Segmentation as a Clinical Computing Problem

Cardiac Magnetic Resonance (CMR) imaging is widely used for non-invasive assessment of cardiac morphology and function. In the ACDC benchmark, the segmentation targets are the Right Ventricle (RV), Myocardium (MYO), and Left Ventricle (LV), which are clinically important because they support the computation of ventricular volumes, myocardial mass, and ejection fraction (Bernard et al., 2018). These quantities are not obtained directly from a neural network classifier; they require spatial masks from which anatomical volumes can be computed. Therefore, segmentation quality is central to the reliability of downstream cardiac function analysis.

CMR segmentation is challenging because the difficulty is not uniform across all structures. LV segmentation is often more stable because the LV cavity is relatively compact and has a clearer blood-pool boundary. In contrast, the myocardium is a thin ring-like structure whose boundaries can be affected by partial volume effects, papillary muscles, and signal ambiguity. The RV is also difficult because it has a more variable crescent-like shape and thinner wall. These anatomical observations explain why a single aggregate Dice score is insufficient; class-specific performance should also be reported when evaluating cardiac segmentation models.

The ACDC dataset is useful for this thesis because it includes multiple cardiac conditions, including normal subjects and pathological groups such as dilated cardiomyopathy, hypertrophic cardiomyopathy, myocardial infarction, and abnormal right ventricle cases (Bernard et al., 2018). This heterogeneity makes the benchmark more relevant than a dataset containing only healthy cardiac anatomy. Nevertheless, the ACDC training cohort remains a limited dataset, so patient-level separation between training and validation is essential. If frames from the same patient appear in both training and evaluation, the reported performance may reflect memorization of patient-specific anatomy rather than generalization.

4.2 Traditional Medical Image Segmentation Methods

Before deep learning became dominant, medical image segmentation commonly relied on intensity thresholding, deformable models, graph-based optimization, and level-set methods. These approaches remain important historically because they clarify why learning-based segmentation became attractive.

A simple thresholding method assumes that the target object can be separated from the background by intensity. Otsu’s method, for example, selects a threshold T by maximizing the between-class variance. If the two classes have probabilities $\omega_0(T)$ and $\omega_1(T)$, and means $\mu_0(T)$ and $\mu_1(T)$, the objective can be written as:

$$\sigma_B^2(T) = \omega_0(T)\omega_1(T) [\mu_0(T) - \mu_1(T)]^2. \quad (4.1)$$

This formulation is elegant because it does not require annotated training data. However, it is not sufficient for cardiac MRI segmentation. MRI intensities are not standardized in the same way as CT Hounsfield Units, and myocardium, blood pool, and surrounding tissues may overlap in intensity. As a result, a global threshold cannot reliably separate

RV, MYO, and LV across patients.

Active contour models, introduced by Kass et al., formulate segmentation as an energy minimization problem (Kass et al., 1988). A contour evolves under internal smoothness forces and external image forces. A simplified energy functional can be written as:

$$E_{snake} = \int_0^1 \left[\frac{1}{2} \alpha |v'(s)|^2 + \frac{1}{2} \beta |v''(s)|^2 + E_{image}(v(s)) \right] ds. \quad (4.2)$$

Here, the first two terms regulate contour continuity and smoothness, while E_{image} attracts the contour to image features such as edges. This is useful when boundaries are clear, but cardiac MRI boundaries can be weak or ambiguous at basal and apical slices. Moreover, deformable models often require careful initialization, which limits their practicality for fully automated pipelines.

Level-set methods represent the contour implicitly as the zero level of a higher-dimensional function ϕ (Osher & Sethian, 1988). A typical evolution equation is:

$$\frac{\partial \phi}{\partial t} + F|\nabla \phi| = 0. \quad (4.3)$$

The implicit representation allows topology changes more naturally than parametric snakes. However, iterative numerical optimization and sensitivity to image forces remain practical limitations. For this reason, traditional methods are useful background, but they do not remove the need for robust data-driven segmentation in heterogeneous CMR datasets (Peng et al., 2020).

4.3 Deep Learning Segmentation Architectures

4.3.1 Fully Convolutional Networks and U-Net

Fully Convolutional Networks (FCNs) showed that classification-style neural networks could be reformulated for dense pixel-wise prediction by replacing fully connected layers with convolutional operations (Long et al., 2015). This idea was important because segmentation requires a prediction for every pixel or voxel rather than one image-level label.

U-Net became one of the most influential architectures in biomedical image segmentation (Ronneberger et al., 2015). Its main contribution is the encoder-decoder structure with skip connections. The encoder gradually reduces spatial resolution to learn semantic features, while the decoder restores resolution to produce a segmentation mask. Skip connections transfer high-resolution spatial features from the encoder to the decoder, which helps recover fine boundaries. If $\mathbf{X}_{enc}^{(l)}$ is an encoder feature map and $\mathbf{X}_{dec}^{(l+1)}$ is a decoder feature map from a coarser level, a simplified skip-connection operation can be written as:

$$\mathbf{X}_{dec}^{(l)} = \mathcal{F} \left(\mathbf{X}_{enc}^{(l)} \oplus \mathcal{U} \left(\mathbf{X}_{dec}^{(l+1)} \right) \right), \quad (4.4)$$

where $\mathcal{U}$ denotes up-sampling, $\oplus$ denotes channel-wise concatenation, and $\mathcal{F}$ denotes convolutional refinement. This equation explains why U-Net is suitable for medical segmentation: the decoder does not reconstruct boundaries only from compressed semantic features; it also receives high-resolution information from earlier encoder layers.

For volumetric data, 3D U-Net extends the same principle from 2D images to 3D tensors (Çiçek et al., 2016). V-Net also used volumetric convolution and Dice-based optimization for 3D medical segmentation (Milletari et al., 2016). These works are directly relevant to this thesis because the proposed Nano-Mamba U-Net uses a 3D U-Net-style encoder-decoder as its architectural base. The contribution of this thesis is not the invention of the U-Net framework itself. Rather, the project studies whether the bottleneck of

such a framework can be modified into a compact sequence-processing module.

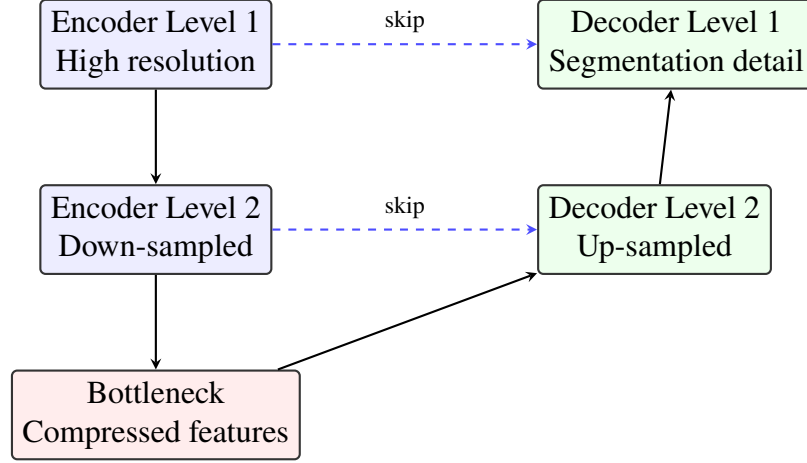


Figure 4.1: Conceptual encoder-decoder structure of U-Net-style segmentation models. Encoder blocks learn increasingly abstract features, decoder blocks recover spatial resolution, and skip connections preserve boundary information.

4.3.2 Attention U-Net

Attention U-Net introduced attention gates to suppress irrelevant encoder features before they are passed through skip connections (Oktay et al., 2018). This is relevant in medical images because large regions of an image may contain background structures unrelated to the target anatomy. In a simplified attention gate, an encoder feature x and a decoder gating signal g are projected into a shared representation and transformed into an attention coefficient:

$$\alpha = \sigma \left(\psi^T \delta (W_x x + W_g g + b) \right). \quad (4.5)$$

The filtered feature can then be represented as:

$$\hat{x} = \alpha \odot x. \quad (4.6)$$

Here, σ is a sigmoid function, δ is a nonlinear activation, and $\odot$ denotes element-wise multiplication. The key idea is not that attention automatically guarantees better segmentation, but that it provides a mechanism for feature selection. In the final experiment of

this thesis, Attention U-Net did not outperform the other models, which illustrates why empirical comparison under the same split is necessary.

4.3.3 Residual Networks and SegResNet

Residual learning was introduced to make deeper networks easier to optimize (He et al., 2016). Instead of asking stacked layers to learn a full mapping $\mathcal{H}(x)$, a residual block learns a residual function $\mathcal{F}(x)$ and adds the input back to the output:

$$y = \mathcal{F}(x) + x. \quad (4.7)$$

This identity shortcut improves gradient flow and can make deeper convolutional models more trainable. SegResNet-style medical segmentation architectures use residual blocks in an encoder-decoder setting and have become strong baselines in medical image segmentation practice. In this thesis, SegResNet16 serves as an important comparison model because it represents a strong residual convolutional alternative to the proposed Nano-Mamba U-Net. The final results show that SegResNet16 achieved the highest mean Dice, which must be acknowledged rather than hidden.

4.4 Transformers and Sequence Modeling in Medical Imaging

Transformers introduced self-attention as a way to model long-range dependencies (Vaswani et al., 2017). Vision Transformer adapted this idea to image classification by treating image patches as tokens (Dosovitskiy et al., 2020). In medical imaging, UNETR used transformer encoders for 3D segmentation (Hatamizadeh, Tang, et al., 2022), while Swin Transformer and Swin UNETR introduced hierarchical shifted-window attention to reduce the cost of global attention (Hatamizadeh, Nath, et al., 2022; Liu et al., 2021).

The computational concern with standard self-attention is that the attention matrix

grows quadratically with the number of tokens. For an input sequence of length N , the attention matrix has size $N \times N$, and the simplified attention operation is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V. \quad (4.8)$$

This equation shows why sequence length matters: the multiplication QK^T compares every token with every other token. Window-based methods reduce this cost, but the broader issue remains important for volumetric segmentation because 3D images can produce many spatial tokens.

It would be inaccurate to claim that all transformer models are infeasible or that they are unsuitable for medical imaging. Prior work shows that transformer-based segmentation can be effective. The more careful motivation for this thesis is narrower: for a master's project focused on lightweight deployment and reproducible experiments on a consumer GPU, it is reasonable to investigate alternatives that attempt to combine broader context modeling with lower parameter count.

4.5 State Space Models and Mamba-Inspired Design

State Space Models (SSMs) represent sequence dynamics through a hidden state. A continuous-time linear state space model can be written as:

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t). \quad (4.9)$$

Here, $h(t)$ is the hidden state, $x(t)$ is the input signal, $y(t)$ is the output signal, and A , B , and C are state transition and projection matrices. In the context of deep learning, the appeal of SSMs is that they can model long sequences without explicitly constructing a full $N \times N$ attention matrix.

Mamba extends this line of work through selective state space modeling and hardware-aware sequence processing (Gu & Dao, 2023). Recent biomedical segmentation studies have also explored Mamba-style modules for medical image segmentation, including U-Mamba for long-range dependency modeling and SegMamba for 3D medical segmentation (Ma et al., 2024; Xing et al., 2024). These works support the broader relevance of SSM-inspired modeling in medical image analysis, but they should be treated as related work rather than as evidence that every Mamba-inspired block will automatically improve cardiac segmentation.

However, this thesis must make a clear distinction between the original Mamba implementation, recent Mamba-based medical segmentation studies, and the implemented project model. The proposed Nano-Mamba U-Net uses a Mamba-inspired bottleneck implemented in PyTorch. It serializes compressed 3D feature maps into a sequence and applies a lightweight gated sequence-processing block. It is therefore more accurate to describe the contribution as a Mamba-inspired architectural exploration, not as a full reproduction of the original selective scan mechanism.

This distinction matters for scientific integrity. If the implementation does not include the full selective scan algorithm, the thesis should not claim that it has implemented the complete Mamba architecture. The contribution is still valid, but it should be framed correctly: the project investigates whether a compact sequence-inspired bottleneck can provide a useful accuracy-efficiency trade-off in cardiac MRI segmentation.

4.6 Loss Functions and Evaluation Metrics

Medical segmentation is often class-imbalanced because background voxels dominate the image volume, while structures such as myocardium occupy a smaller region. Dice-based objectives are commonly used because they directly optimize overlap between prediction and ground truth (Milletari et al., 2016; Sudre et al., 2017). For a predicted binary

mask P and ground-truth mask G , the Dice Similarity Coefficient is:

$$\text{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \quad (4.10)$$

In multi-class segmentation, Dice can be computed separately for RV, MYO, and LV, and then averaged. This is the reporting strategy used in this thesis.

Cross-entropy is also widely used because it provides voxel-wise classification supervision. If y_i is the true class at voxel i and $p_{i,c}$ is the predicted probability for class c , the multi-class cross-entropy can be expressed as:

$$\mathcal{L}_{CE} = - \sum_i \sum_c \mathbf{1}(y_i = c) \log p_{i,c}. \quad (4.11)$$

The final experiment uses MONAI’s Dice-Cross Entropy loss, combining region-overlap supervision with voxel-wise classification supervision. This choice is practical rather than novel; it follows established segmentation practice and is supported by the MONAI framework (Cardoso et al., 2022).

Evaluation should also avoid leakage. If the same patient contributes samples to both training and validation, the measured Dice may overestimate generalization. Therefore, the final experimental design of this thesis uses patient-level splitting. This methodological point is as important as the architecture itself, because a strong model comparison is only meaningful when all models are trained and evaluated under the same split.

4.7 Summary of the Literature Gap

The literature shows that U-Net-style models remain strong foundations for biomedical segmentation, attention mechanisms can improve feature selection, residual models provide strong convolutional baselines, and transformer/SSM approaches offer ways to

model broader contextual relationships. Recent Mamba-style medical segmentation studies further suggest that sequence and state-space ideas are becoming relevant in this area. The gap addressed by this thesis is not the absence of segmentation models, but the need to study a compact architecture under a rigorous and reproducible validation protocol.

Based on this review, the proposed Nano-Mamba U-Net is positioned as a lightweight Mamba-inspired extension of a 3D U-Net-style segmentation model. The appropriate research question is therefore not whether it universally surpasses all previous methods, but whether it offers a useful accuracy-efficiency trade-off for cardiac MRI segmentation on the ACDC dataset.

CHAPTER 5: RESEARCH DESIGN AND METHODOLOGY

This chapter describes the research design used to implement and evaluate the proposed Nano-Mamba U-Net. The methodology is written to make the experiment reproducible: it specifies the dataset boundary, data discovery procedure, patient-level split, preprocessing pipeline, model architecture, loss function, checkpoint selection, evaluation metrics, and computational profiling. The chapter also distinguishes clearly between methods adopted from prior work and components implemented in this project.

5.1 Overall Research Workflow

The study follows an experimental computing workflow. First, the ACDC cardiac MRI data are discovered and paired with corresponding segmentation labels. Second, patients are separated into training and validation groups using a deterministic patient-level split. Third, all models are trained under the same preprocessing, optimization, and validation protocol. Finally, the best checkpoint for each model is selected by validation mean Dice and evaluated on the held-out validation patients.

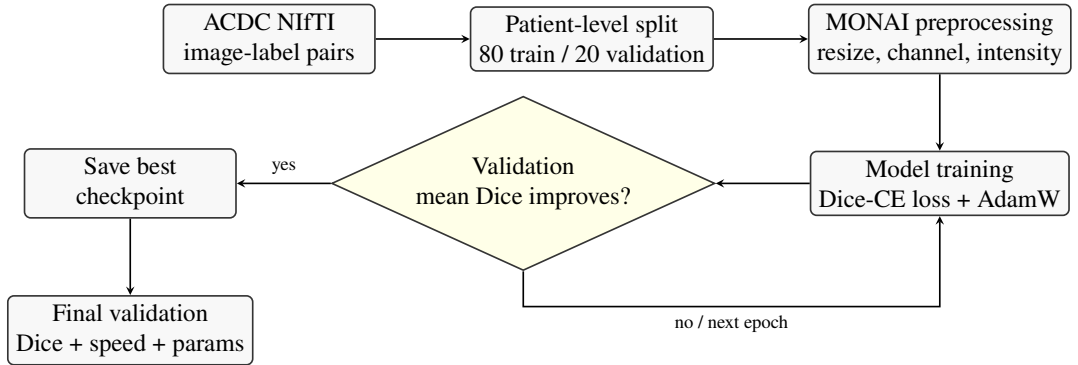


Figure 5.1: Reproducible experimental workflow. The main methodological safeguard is patient-level separation before training, so that validation cases come from patients not seen during optimization.

5.2 Dataset and Data Discovery

The experiment uses the Automated Cardiac Diagnosis Challenge (ACDC) dataset (Bernard et al., 2018). The task is multi-class segmentation of three anatomical structures: RV, MYO, and LV. The final experiment discovered 200 valid image-label pairs from 100 patients. Each patient contributes labelled end-diastolic and end-systolic frames, giving two labelled 3D cases per patient.

A practical issue in this project is that the local dataset layout contains paths ending in .nii that may act as containers rather than directly readable NIfTI files. Therefore, the final data discovery procedure checks whether each candidate path is a valid non-empty file or a directory containing the actual NIfTI file. Empty placeholders are skipped. This step is important because directly passing an empty placeholder file to a NIfTI reader causes a loading error. The final data discovery report confirmed 200 valid image-label pairs and no skipped invalid containers.

5.3 Patient-Level Training and Validation Split

The split was performed at patient level rather than case level. This means all labelled frames from the same patient belong either to training or to validation, never both. This design prevents leakage of patient-specific anatomy into the validation set. With random seed 42 and an 80/20 split, the experiment used 80 training patients and 20 validation patients, corresponding to 160 training cases and 40 validation cases.

Let $\mathcal{P}$ denote the set of all patient identifiers. The split can be expressed as:

$$\mathcal{P}_{train} \cap \mathcal{P}_{val} = \emptyset, \quad \mathcal{P}_{train} \cup \mathcal{P}_{val} = \mathcal{P}. \quad (5.1)$$

This condition is the core methodological safeguard of the experiment. It ensures that the validation Dice measures performance on unseen patients rather than on additional frames

from patients used during training.

5.4 Preprocessing Pipeline

All models use the same preprocessing pipeline implemented with MONAI (Cardoso et al., 2022). Each image and label pair is loaded from NIfTI format, converted to channel-first representation, resized to a fixed spatial size of $256 \times 256 \times 16$, intensity-scaled, and converted to a PyTorch tensor. Applying the same preprocessing to all models is necessary because otherwise observed performance differences could be caused by data handling rather than architecture.

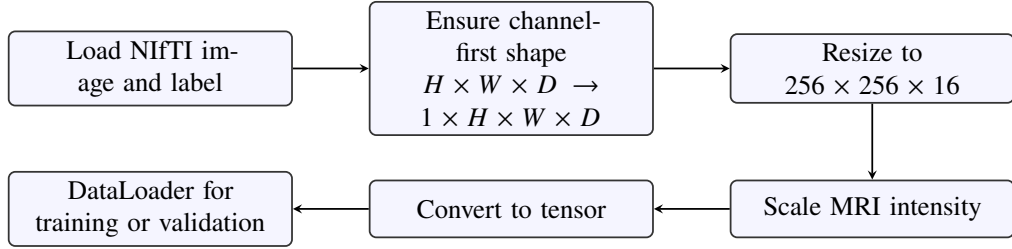


Figure 5.2: Preprocessing pipeline used for all compared models. Images are interpolated continuously, while labels are resized with nearest-neighbour interpolation to preserve class identities.

For continuous MRI intensity volumes, trilinear interpolation is used during resizing. If $V(x, y, z)$ denotes the interpolated intensity at a continuous coordinate, the interpolation can be described as a weighted combination of its eight neighbouring voxels:

$$V(x, y, z) \approx \sum_{i=0}^1 \sum_{j=0}^1 \sum_{k=0}^1 w_{ijk} V_{ijk}. \quad (5.2)$$

Here, V_{ijk} are neighbouring voxel intensities and w_{ijk} are interpolation weights determined by the relative distance between the target coordinate and the neighbouring grid points. For segmentation labels, nearest-neighbour interpolation is used instead, because labels are categorical class identifiers rather than continuous intensity values.

MRI intensity values are relative and scanner-dependent rather than standardized physical units (Nyúl & Udupa, 1999). Therefore, intensity scaling is applied before network input. A simple min-max formulation is:

$$S(I_{xyz}) = \frac{I_{xyz} - \min(I)}{\max(I) - \min(I) + \epsilon}. \quad (5.3)$$

The constant ϵ prevents division by zero. In practice, MONAI handles these operations in the transform pipeline. The mathematical expression is included to clarify why intensity normalization is needed for stable neural network training.

5.5 Nano-Mamba U-Net Architecture

The proposed model follows a 3D U-Net-style encoder-decoder structure. The encoder uses convolutional blocks and pooling to reduce spatial resolution while increasing feature channels. The decoder uses transposed convolutions and skip connections to recover spatial resolution and preserve boundary information. The architectural modification studied in this thesis is the insertion of a lightweight Mamba-inspired bottleneck at the compressed latent representation.

Each convolutional block consists of two 3D convolutional layers followed by batch normalization and ReLU activation. A convolutional layer transforms a local neighbourhood into an output feature using learned kernels. For a 3D convolution, this can be written as:

$$Y_{c_o}(p) = \sum_{c_i} \sum_{q \in \mathcal{K}} W_{c_o, c_i}(q) X_{c_i}(p + q) + b_{c_o}. \quad (5.4)$$

Here, X is the input feature map, Y is the output feature map, $\mathcal{K}$ is the local 3D kernel support, and W and b are trainable parameters. This equation also shows the locality of convolution: each output voxel depends on a neighbourhood around position p .

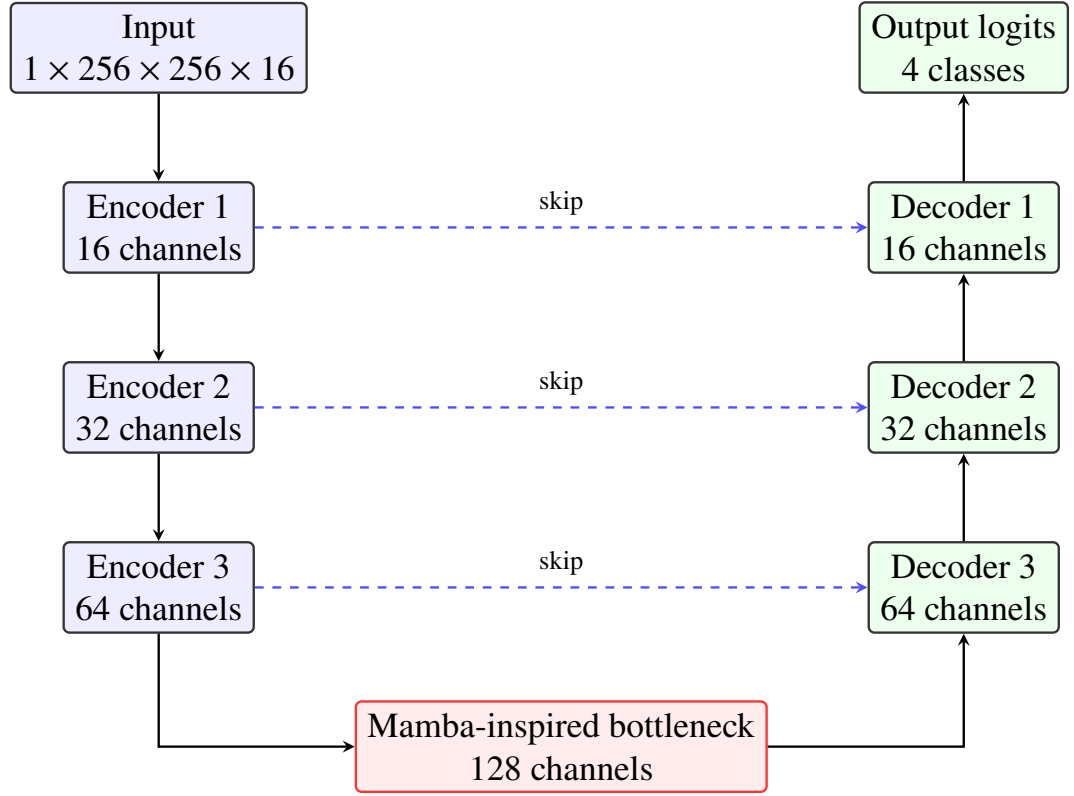


Figure 5.3: Nano-Mamba U-Net architecture. The model keeps the U-Net encoder-decoder and skip-connection structure, while replacing the deepest bottleneck with a compact Mamba-inspired sequence-processing module.

Batch normalization stabilizes training by normalizing intermediate features within a mini-batch (Ioffe & Szegedy, 2015). For a feature value x , batch normalization can be expressed as:

$$\hat{x} = \frac{x - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}, \quad y = \gamma\hat{x} + \beta. \quad (5.5)$$

The parameters γ and β are learned, allowing the network to recover suitable feature scales after normalization. ReLU activation is then applied as:

$$f(x) = \max(0, x). \quad (5.6)$$

These components are standard neural network operations and are not claimed as novel contributions of this thesis.

5.6 Mamba-Inspired Bottleneck Implementation

The bottleneck receives a compressed 3D feature map. Let this tensor be:

$$\mathcal{X} \in \mathbb{R}^{B \times C \times H \times W \times D}. \quad (5.7)$$

The tensor is serialized across the spatial dimensions to obtain a sequence:

$$\mathbf{X}_{seq} = \text{reshape}(\mathcal{X}) \in \mathbb{R}^{B \times L \times C}, \quad L = HWD. \quad (5.8)$$

This operation does not create new anatomical information. It changes the representation so that the bottleneck can process compressed volumetric features as a sequence.

The implemented bottleneck is Mamba-inspired rather than a full selective scan implementation. It uses linear projection, depthwise one-dimensional convolution, gating, output projection, and residual addition. A simplified representation of the operation is:

$$\mathbf{Z} = \text{Proj}_{out} (\text{Gate} (\text{DWConv1D} (\text{Proj}_{in} (\mathbf{X}_{seq})))) + \mathbf{X}_{seq}. \quad (5.9)$$

Here, Proj_{in} and Proj_{out} are learned linear projections, DWConv1D is a depthwise sequence convolution, and Gate denotes a learned nonlinear gating operation. The output is reshaped back into 3D feature form before being passed to the decoder.

This design choice must be interpreted carefully. The original Mamba paper proposes selective state space sequence modeling (Gu & Dao, 2023). The implementation in this project uses a lightweight PyTorch block inspired by this direction, but it does not claim to reproduce the full hardware-aware selective scan. The research contribution is therefore the integration and evaluation of a compact sequence-inspired bottleneck inside a cardiac MRI segmentation model.

5.7 Comparison Models and Ablation Design

The experiment compares Nano-Mamba U-Net with several reference architectures. The 3D U-Net baseline represents the standard encoder-decoder approach. Attention U-Net represents attention-gated skip connections. SegResNet16 represents a residual convolutional segmentation model. Two ablation variants are included to study the role of the bottleneck: No-Mamba U-Net replaces the Mamba-inspired bottleneck with convolutional blocks, while Half-Mamba U-Net reduces the bottleneck capacity.

The purpose of the ablation study is not to prove that every Mamba-inspired configuration must outperform every convolutional configuration. Instead, the ablation tests whether the proposed bottleneck improves the accuracy-efficiency trade-off under the same training and validation protocol. This distinction is important because the final results show that No-Mamba U-Net slightly outperformed Nano-Mamba U-Net in mean Dice, while Nano-Mamba U-Net retained the lowest parameter count.

5.8 Training Objective and Optimization

The models are trained using a Dice-Cross Entropy objective. Dice loss addresses region-level overlap, while cross entropy provides voxel-wise class supervision. For class c , the Dice score is:

$$\text{DSC}_c = \frac{2 \sum_i p_{i,c} g_{i,c}}{\sum_i p_{i,c} + \sum_i g_{i,c} + \epsilon}. \quad (5.10)$$

The corresponding Dice loss can be written as:

$$\mathcal{L}_{Dice} = 1 - \frac{1}{C} \sum_{c=1}^C \text{DSC}_c. \quad (5.11)$$

The multi-class cross-entropy term is:

$$\mathcal{L}_{CE} = - \sum_i \sum_c g_{i,c} \log(p_{i,c}). \quad (5.12)$$

The total loss used for optimization is:

$$\mathcal{L}_{total} = \mathcal{L}_{Dice} + \mathcal{L}_{CE}. \quad (5.13)$$

Here, $p_{i,c}$ denotes the predicted probability for voxel i and class c , while $g_{i,c}$ denotes the one-hot ground-truth label. This loss is a standard practical choice for medical segmentation and is implemented using MONAI’s Dice-Cross Entropy loss.

Optimization is performed with AdamW (Loshchilov & Hutter, 2017). The learning rate is 1×10^{-4} and weight decay is 1×10^{-5} . All models are trained for 150 epochs. The batch size is set according to model memory requirements: lighter models use batch size 2, while heavier models use batch size 1. These settings are kept consistent in the final experimental script to support fair comparison.

5.9 Checkpoint Selection and Evaluation Metrics

A key methodological correction in the final experiment is validation-based checkpoint selection. Earlier exploratory scripts saved best models according to training loss, which is not a reliable indicator of generalization. In the final experiment, the checkpoint is saved only when validation mean Dice improves. If $\overline{D}_{val}^{(e)}$ denotes the validation mean Dice at epoch e , the selected checkpoint is:

$$e^* = \arg \max_e \overline{D}_{val}^{(e)}. \quad (5.14)$$

The final reported metrics are computed from this selected checkpoint on the held-out validation cases.

The primary reported segmentation metric is mean Dice across RV, MYO, and LV:

$$\overline{D} = \frac{D_{RV} + D_{MYO} + D_{LV}}{3}. \quad (5.15)$$

The study also reports parameter count and inference speed. Parameter count measures the number of trainable parameters in the model, while inference speed is measured as frames per second on the same hardware environment. These computational metrics are necessary because the research question concerns the trade-off between accuracy and lightweight design, not Dice score alone.

5.10 Hardware and Reproducibility

All experiments are run using PyTorch and MONAI on a single NVIDIA RTX 4060 GPU with 8GB VRAM. The random seed is fixed to 42 for Python, NumPy, and PyTorch. CuDNN deterministic settings are enabled to improve reproducibility. The final experiment script saves the patient split, per-case validation scores, training logs, checkpoints, and summary CSV/JSON files. These outputs provide an auditable evidence chain linking the reported thesis tables to the executed code.

5.11 Methodological Summary

The methodology is designed to support a fair and cautious evaluation. The proposed Nano-Mamba U-Net is compared with several baseline and ablation models under the same patient-level split and preprocessing pipeline. The model is not evaluated using training-set performance, and it is not claimed to be a full implementation of the original Mamba selective scan. Instead, the experiment tests whether a compact Mamba-inspired

bottleneck can provide competitive cardiac MRI segmentation performance with a low parameter count.

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